Troy Mayer

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EDUCATION

Purdue University, School of Mechanical Engineering and School of Science **West Lafayette, IN**
B.S. Mechanical Engineering; B.S. Applied Statistics; Minor in Economics **Expected May 2028**
Clubs: Club Swimming, House Band **GPA 3.32/4**

PROFESSIONAL EXPERIENCE

Boston Dynamics **Waltham, MA**
Sensors and Perception Co-Op – Atlas **July 2026 – Present**

- Mapping impacts onto Atlas's mesh with encoders and forward kinematics, estimating impact energy from Khatib's operational-space effective mass and an OLS velocity fit; building a physics-based MuJoCo impact simulation.
- Building an impact forensics pipeline and Flask/Plotly dashboard on a Linux VM over ~28k impact events from ~900 test logs: ETL into SQLite, a 3D impact heatmap, and per-event waveforms used by engineers during testing.
- Modeling hardware thermal behavior in Python, using Savitzky-Golay filtering and multi-start least squares to predict steady-state temperatures across hundreds of units without full-length tests.

Caterpillar (via Purdue Data Mine) **West Lafayette, IN**
Machine Learning Researcher **January 2026 – May 2026**

- Built LSTM/XGBoost forecasts of daily paint use with Monte Carlo intervals, achieving a 4.7% material reduction.

Chortos Lab – Soft Adaptive Systems and Structures **West Lafayette, IN**
Undergraduate Researcher **August 2025 – May 2026**

- Wrote a Python computer-vision tracker and LabVIEW/MATLAB data acquisition to measure gel samples, and used them to formulate gels with over 50% more force output.

Emerson – Discrete Automation (ASCO) **Florham Park, NJ**
Mechanical and Manufacturing Engineer Intern **June 2025 – August 2025**

- Designed and CNC-machined a valve spring canister in Creo with GD&T drawings; ran UL 429 solenoid valve tests.

PROJECTS

Legged Robot State Estimation and Control
Personal Project – *github.com/tmayer2112-ui/Troy_Mayer* **August 2026 – October 2026**

- Built a contact-aided error-state Kalman filter in C++ for a simulated Unitree Go1, fusing IMU and leg kinematics to 0.013 m/s velocity error, against 1.38 m/s for the IMU alone.
- Verified it with GoogleTest (finite-difference Jacobian checks, parity with a Python reference) and an observability analysis of what the sensors can and can't see.
- Stress-tested it under foot slip and contact latency, and fixed the latency and weak-tilt-observability failures.
- Closed the loop in Python simulation with no falls, and added an MIT Cheetah 3-style QP balance controller.

Purdue ASME – Biomechanics **West Lafayette, IN**
Active Orthosis – Electronics Team Lead **April 2024 – May 2026**

- Designed a 1 kHz PI motor velocity controller in C++ for an ESP32-S3, tuned from a motor model with feed-forward and anti-windup, and verified it in co-simulation with a model of the motor and forearm.
- Wrote the FreeRTOS firmware with fault monitors, a watchdog, and a kill switch, tested with GoogleTest in CI.
- Cut simulated EMG-to-motion latency from 103 to 43 ms on real EMG via new signal processing and loop tuning.
- Led the 5-engineer electronics team and redesigned the EMG board schematic in KiCad, raising cross-day gesture decoding from 31% to 87% in emulation.

Asteroid Hopping Robot – Independent Research
Purdue University & Universidad Carlos III de Madrid **October 2026 – Present**

- Building a 3-legged hopping robot and landing rig; designing push-off, landing, and attitude control in MuJoCo.

SKILLS & INTERESTS

- **Tools:** C++, Python, MATLAB, SQL, Linux, Git, CMake, GoogleTest, GitHub Actions CI, Eigen, MuJoCo, FreeRTOS, KiCad, LabVIEW, CAD (CATIA, NX, Fusion, Creo), FEA
- **Methods:** state estimation (Kalman filtering), PI and feed-forward control, QP force control, rigid-body kinematics
- **Languages:** English (Native), Turkish (Fluent), Spanish (Fluent - New Jersey Seal of Biliteracy)
- **Hobbies:** Lifting, Hiking, Guitar, Saxophone